

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

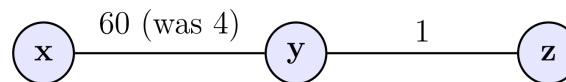
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4$, $c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1 , t_2 , t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?



Solution:

1. Calculations:

- **Initial (t_0):** $D_y(x) = 4$, $D_z(x) = 5$ (via y).
- **t_1 (y detects change):** y updates its distance to x.

$$D_y(x) = \min(c(y, x) + D_x(x), c(y, z) + D_z(x)) = \min(60 + 0, 1 + 5) = 6$$

(Note: y chooses path via z because $1 + 5 < 60$. z still thinks dist to x is 5).

- **t_2 (z receives update from y):** z sees $D_y(x) = 6$.

$$D_z(x) = \min(c(z, y) + D_y(x)) = 1 + 6 = 7$$

- **t_3 (y receives update from z):** y sees $D_z(x) = 7$.

$$D_y(x) = \min(60, 1 + 7) = 8$$

2. Problem & Prevention:

- **Problem: Count-to-Infinity.** The nodes bounce updates back and forth, slowly incrementing the cost until it reaches the direct link cost (60) or infinity.
- **Prevention: Poisoned Reverse.** If z routes to x via y, z should advertise $D_z(x) = \infty$ to y.

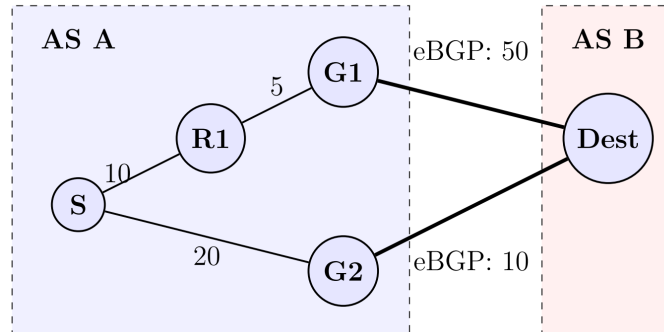
Q2. Carefully read the Question before attempting:

[5 marks]

AS A needs to send a packet to the destination IP in AS B. AS A has two gateways (G1, G2) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?



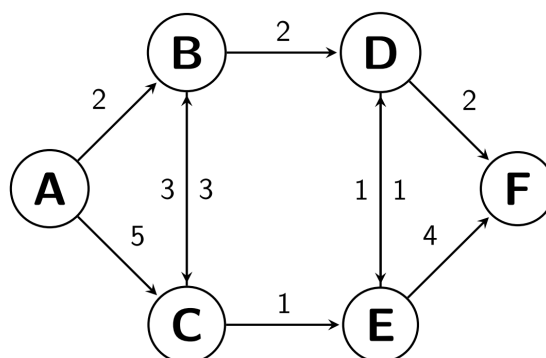
Solution:

- **Internal Costs:** Path to G1 ($S \rightarrow R1 \rightarrow G1$) = $10 + 5 = 15$. Path to G2 ($S \rightarrow G2$) = 20.
- **Hot Potato (Minimize Internal Cost):**
 - Compares internal cost 15 (via G1) vs 20 (via G2). Chooses G1.
 - **Path:** $S \rightarrow R1 \rightarrow G1 \rightarrow \text{Dest}$.
 - **Total Cost:** $15(\text{Intra}) + 50(\text{Inter}) = 65$.
- **Cold Potato (Minimize Total Cost):**
 - Via G1: $15 + 50 = 65$.
 - Via G2: $20 + 10 = 30$.
 - Chooses G2 (since $30 < 65$).
 - **Path:** $S \rightarrow G2 \rightarrow \text{Dest}$.
 - **Total Cost:** 30.
- **Efficiency Penalty:** $\text{Cost}(\text{Hot}) - \text{Cost}(\text{Cold}) = 65 - 30 = 35$.

Q3. Carefully read the Question before attempting:

[5 marks]

Compute the Forwarding Table for **Node E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at **Node E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).



Solution: Running Dijkstra from Source E:

1. **Initialization:** $D(E) = 0$. Neighbors: $D(D) = 1, D(F) = 4$. All others ∞ .
2. **Step 1:** Select node with min dist: **D** (dist=1).
3. **Step 2:** Relax neighbors of D (D goes to E, F).
 - Path to F via D: $D(D) + c(D, F) = 1 + 2 = 3$.
 - Since $3 < 4$ (current $D(F)$), update $D(F)$ to 3. Predecessor of F becomes D.
4. **Step 3:** Select node with min dist: **F** (dist=3). No outgoing edges.
5. Nodes A, B, C are unreachable from E in this directed graph.

Forwarding Table for E:

Destination	Next Hop	Cost
D	D	1
F	D	3
A	- (Unreachable)	∞
B	- (Unreachable)	∞
C	- (Unreachable)	∞